Team Details

Team Name: BYTE BRIGADE

SR. NO	ROLE	NAME	ACADEMIC YEAR
1	**Team Leader**	*Prakhar Bajaj*	*2nd Year*
2	Member 1	*Raghav Soni*	*2nd Year*
3	Member 2	*Parth Kumar*	*2nd Year*
4	Member 3	*Mayank Lodhi*	*2nd Year*

Problem Statement PS01 (KLA) - AI-Based Restoration of Degraded Images for Semiconductor Inspection

COLLEGE NAME

LNCT University, Bhopal

TEAM LEADER CONTACT NUMBER

+91 91796 86667

TEAM LEADER EMAIL ADDRESS

prakharbajaj1551@gmail.com

Problem Statement

PS01 - KLA: AI-Based Restoration of Degraded Images for Semiconductor Inspection

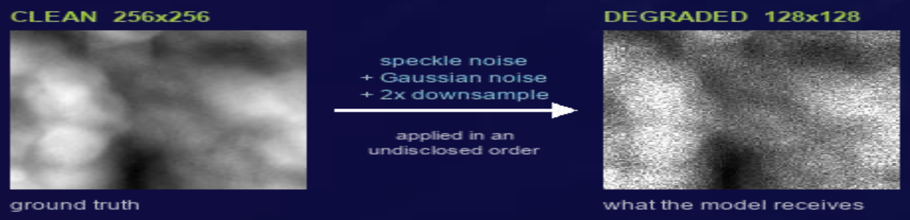
DESCRIPTION / DETAILS

Wafer inspection images arrive noisy and low-resolution: fast, low-dose scanning trades image quality for throughput.

Three degradations are present - speckle noise (scales with brightness), additive Gaussian sensor noise, and 2x downsampling - applied in an undisclosed order.

Real defects hide inside this noise. A missed defect can cost an entire wafer; a false alarm wastes engineer time.

Task: learn a mapping from a 128x128 degraded image to its clean 256x256 original, scored on PSNR, SSIM, LPIPS and end-to-end speed.



Idea Description

One small network restores a noisy 128x128 image to a clean 256x256 one - denoising and 2x super-resolution in a single pass, trained only on the data provided.

KEY CONCEPT & APPROACH

1. *Supervised on the 3,200 provided pairs - no external datasets, no pretrained restoration weights.*
2. *One network performs denoising and 2x super-resolution in a single pass.*
3. *Residual learning - the model corrects rather than generates, which strongly constrains hallucination.*

SOLUTION OVERVIEW

NAFNet-SR: 0.53 M parameters, 2.3 MB checkpoint - runs on a laptop CPU.

Trained on the judged metrics: Charbonnier + SSIM + LPIPS loss.

28.00 dB PSNR / 0.759 SSIM / 0.175 LPIPS - beats bicubic on all three.

End-to-end 25 ms per image on a T4 GPU (40 images/second).

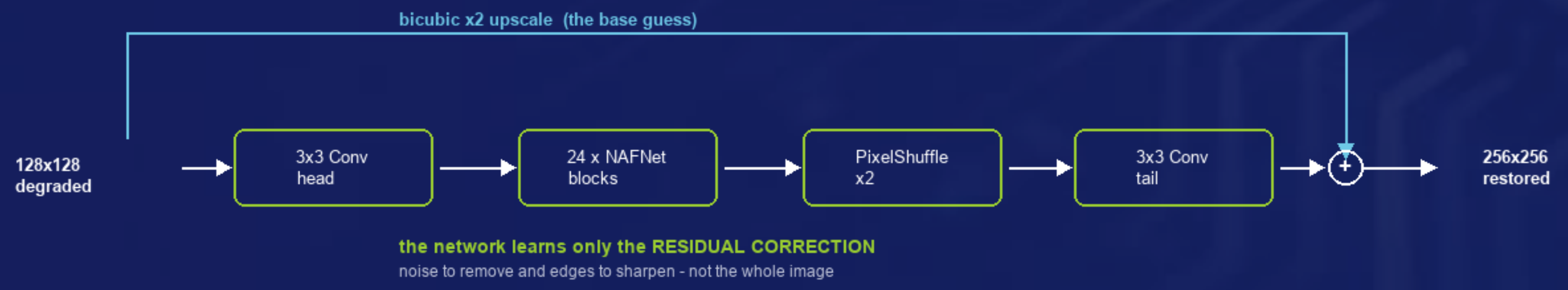
Proposed Solution

NAFNet-SR: 24 NAFNet blocks with 48 channels and a learned 2x PixelShuffle upsampler - 0.53 M parameters, 2.3 MB inference checkpoint.

SOLUTION DETAILS

WHY NAFNet - SOTA quality, no transformer, no activation function.   LOSS - Charbonnier + 0.15 SSIM + 0.05 LPIPS (no GAN loss).

TRAINING - 3,000/100/100 seeded split, 128 px crops, batch 16, 100 epochs, ~2 h on a free Colab T4.   INFERENCE - directories in/out, batched, tiled.



Innovation and Uniqueness

We train directly on the judged metrics, and constrain the model to correct rather than generate - decisive when invented structure could be mistaken for a defect.

KEY INNOVATION

We train directly on the scoreboard: SSIM and LPIPS terms sit inside the loss, so the model optimises two of the three judged metrics rather than hoping they follow from pixel accuracy.

Hallucination is strongly constrained: the global bicubic skip means the network predicts a correction rather than generating an image freely.

Float-faithful pipeline: degraded inputs legitimately exceed [0,1] and we never clip them; clipping is applied only to outputs, where ground truth guarantees the range.

COMPETITIVE ADVANTAGE

SMALLER: 0.53 M parameters (2.3 MB) vs 10-26 M for Restormer or SwinIR - throughput is a strength, not a compromise.

FASTER: 25 ms per image end-to-end on a T4; all 400 test images in ~10 s.

TRUSTWORTHY: beats the bicubic baseline on 100 of 100 held-out images; failure cases are published and analysed rather than hidden.

REPRODUCIBLE: seeded splits, one-command training and inference, weights and runtime report committed to the repository.

KNOWN LIMITATION: very fine texture gains least - downsampling destroys detail we will not invent.

Impact and Benefits

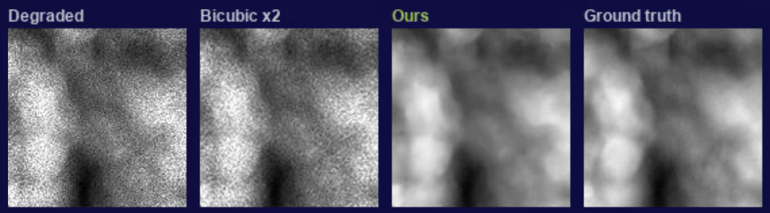
Better on all three judged image-quality metrics, at 25 ms per image - recovering detail that noise would otherwise hide, with no new hardware.

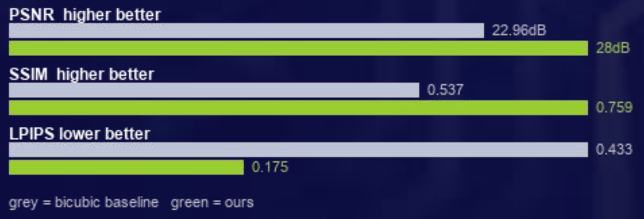
Primary Impact

Detail hidden inside sensor noise becomes visible again - fewer missed defects and fewer false alarms.
Enables faster or lower-dose scanning: recover quality in software, so existing microscopes inspect more wafers.

Quantifiable Outcomes

PSNR 22.96 -> 28.00 dB (+5.04 dB)
SSIM 0.537 -> 0.759 | LPIPS 0.433 -> 0.175
25 ms/image - all 400 test images in ~10 s
0.53 M parameters, 2.3 MB checkpoint - runs on CPU





Technology & Feasibility/Methodology Used

PyTorch and NAFNet-SR, trained in about two hours on a free Colab T4, and deployable on either a GPU or a plain laptop CPU.

IMPLEMENTATION STRATEGY

Trained end to end on a free Google Colab T4 in about two hours; no paid infrastructure was used at any stage.
Inference runs on GPU or CPU. Batched on CUDA, tiled for large images, and verified on 128, 256 and non-square inputs.
Deployment is a drop-in folder-to-folder script plus a Streamlit UI for operators; evaluators need no source-code edits.
Measured, not estimated: benchmark.py reports end-to-end runtime with hardware and software versions stated.

PyTorch 2.11 - NAFNet-SR

NVIDIA T4 / H100 - CPU capable

Colab - Git - Streamlit

GitHub & Video Link

GitHub Repository

github.com/prakharbajaj1551/semicon-image-restoration

Prototype / Simulation Video

https://youtu.be/0eGuJqKfNBQ

Research and References

Research Background & Methodology

Every component is drawn from peer-reviewed restoration research, not invented for this hackathon.

Built on published, peer-reviewed research rather than an unproven design: NAFNet (ECCV 2022) matches transformer restorers at a fraction of the cost.
Loss follows restoration literature - Charbonnier for pixel fidelity, SSIM for structure, LPIPS for perceptual quality.
Validated on a held-out split with no selection leakage, against a bicubic baseline, with failure cases analysed.

References & Citations

The three papers behind our architecture, our upsampler and our perceptual metric.

Ref 1: Chen et al., Simple Baselines for Image Restoration (NAFNet), ECCV 2022 - arxiv.org/abs/2204.04676

Ref 2: Zhang et al., The Unreasonable Effectiveness of Deep Features as a Perceptual Metric (LPIPS), CVPR 2018

Ref 3: Shi et al., Real-Time Single Image Super-Resolution Using an Efficient Sub-Pixel CNN, CVPR 2016